

Matthew J. Maxwell

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AOS

Philosophy of Science (biology, historical sciences)

AOC

Philosophy of Artificial Intelligence, Bioethics, Epistemology (Social and Formal), Logic

Education

Ph.D. Philosophy <i>University of Wisconsin-Madison</i>	2027 (expected)
<i>University of Rochester (no degree taken)</i>	2018–2020
Masters of Arts, Philosophy <i>Simon Fraser University</i>	2018
Bachelors of Arts, Philosophy <i>Iowa State University</i>	2015

Dissertation

Philosophical Aspects of Genetic Load (Expected defense date: 2027)
Committee: Elliott Sober (co-chair), Aja Watkins (co-chair), Hayley Clatterbuck, Clinto Castro, John Poole (?), Alex Meehan (?)

Publications

Journal Articles

"Comparing the strengths of two causes of an effect — zeroing-out, adding-in, and the need for baselines"

Synthese, Forthcoming <https://doi.org/10.1007/s11229-025-05435-3> (with Elliot Sober)

"Fitness, Mutational Load, and Eugenics"

Philosophy of Science, 2025 <https://doi.org/10.1017/psa.2025.10137>

"Gradualism, Natural Selection, and the Randomness of Mutation — Fisher, Kimura, and Orr, Connecting the Dots"

Under Review

"When Does Drift Dominate Selection? A Zeroing-Out Criterion." *Philosophy, Theory, and Practice in Biology* (with Elliott Sober)

When selection and drift both causally influence an allele's fixation in a population, which cause had the greater influence? We develop a zeroing-out criterion, according to which you can determine which had a stronger influence by comparing the probabilities of fixation under a selection-only and drift-only condition.

In Progress

"Is Population Structure Arbitrary? STRUCTURE-Like Algorithms and the Search for Disease-Linked Loci." Draft available upon request.

The algorithms STRUCTURE and ADMIXTURE are commonly used to discover population structure, but have been severely criticized. In this paper I argue that for the purpose of finding disease-linked loci, these algorithms do not face the same concerns that they do for other purposes.

"Measuring Fitness: Generality and Other Desiderata." Draft available on request

I argue that there are four desiderata for an account of fitness—generality, ubiquity, uniqueness, and realism. It is not, however, possible to satisfy all four of these desiderata together, thus theorists of fitness must make a choice.

"Fishing for Evidence: Distinguishing Evidence from Observation in Observation Selection Effects." Draft available upon request

Discussions of observation selection effects have ignored the distinction between observation and evidence. Evidence for a hypothesis, I argue, is distinct from the observation of that evidence. Unobservability of evidence does not entail that the evidence does not obtain. What is required for an OSE is that evidence is guaranteed, not that counter-evidence is unobservable.

"What Kind of Law is the Zero-Force Evolutionary Law: Lessons from the Genetic Load Research Program." In preparation.

I argue that Brandon and McShea's (2010) Zero-Force Evolutionary Law is in conflict with the genetic load research program and that this would have hindered advances in genetics, including Kimura's neutral theory, had the ZFEL gestalt shift been taken seriously.

"Precision in the Art of Navigation: a study of significant figures in Norie's *Epitome of Practical Navigation*." Draft available on request

Significant figures, or significant digits, are an understudied aspect of scientific practice. They are not conventional rules for rounding, a statistical concept, or a measure of error. Rather they are a report of the level of resolution of the measurement procedure and play a distinctive role in the measurement, calculation, and reporting of

measured values. As such they instantiate Grice's conversational maxims of quantity and quality. I use the case study of John W. Norie's 1805 *A New and Complete Epitome of Navigation* to explore their use.

Book Reviews

"Veigl and Currie (eds), *Methods in the philosophy of science: a user's guide.*"

Metascience, 2026 <https://doi.org/10.1007/s11016-026-01146-8> with Aja Watkins and Andrew Zeppa

"Noah A. Rosenberg (2025), *Mathematical Properties of Population-Genetic Statistics: Quadratic Forms Most Beautiful.*"

Acta Biotheoretica, 2026 <https://doi.org/10.1007/s10441-026-09517-8>

Professional Presentations

"Is Population Structure Arbitrary? STRUCTURE-Like Algorithms and the Search for Disease-Linked Loci"

Society for Philosophy of Science in Practice, University of Cambridge July 2026

"Must Population Structure be Arbitrary?"

Philosophy and Biology Shop Talks, Westfield, North Carolina June 2026

"What Kind of Law Is the Zero-Force Evolutionary Law? Lessons From the Genetic Load Research Project"

International Society for the History, Philosophy and Social Studies of Biology, Porto, Portugal July 2025

"Measuring Fitness: Generality and Other Desiderata"

Canadian Society for the History and Philosophy of Science, Toronto May 2025

"Fitness, Mutational Load, and Eugenics"

Philosophy of Science Association, New Orleans November 2024

"The Evidential Significance of Model Disagreement"

Society for Philosophy of Science in Practice, University of South Carolina May 2024

"The Evidential Significance of Model Disagreement"

Unity & Disunity in Science, University of Notre Dame (poster) April 2024

"Prospects for a counterfactual account of junk DNA"

International Society for the History Philosophy and Social Studies of Biology Biennial Meeting, University of Toronto July 2023

"Precision in the Art of Navigation: a study of significant figures in Norie's *Epitome of Practical Navigation*"

Canadian Society for the History and Philosophy of Science, Toronto May 2023

"The Evidence-Observation Distinction in Observation Selection Effects"

"Gradualism, Mutation, and Natural Selection—Darwin, Fisher, and Kimura"
International Society for the History, Philosophy and Social Studies of Biology, Cold
Spring Harbor Laboratory, with Elliott Sober July 2021

"Bayesian Sensitivity and the Law"
Western Canadian Philosophical Association, University of Lethbridge October 2019

"Ockham, Miracles, and Causal Regularity"
Johns Hopkins University Philosophy Graduate Student Conference, Johns Hopkins
University October 2019

"Bridging Accuracy and Epistemic Merit"
Pitt-CMU Graduate Student Philosophy Conference, Carnegie Mellon University
March 2019

"Smart Shopping for Inductive Methods"
Western Canadian Philosophical Association, University of Alberta October 2016

"Blocking the Path of Inquiry: Fine and Peirce on Acceptance"
Science and Values in Peirce and Dewey, University of Waterloo April 2016

Honors and Awards

Marcus G. and Blanche L. Singer Graduate Fellowship 2026

Awarded to a dissertator who expect to complete their dissertation the following year.

Mary B. Hesse Graduate Student Essay Award, Honorable Mention 2024

Runner-up for the best single-authored paper submitted by a graduate student as a con-
tributed paper for the *Philosophy of Science Association conference*.

William H. Hay Summer Fellowship 2024

For excellence in a summer research project.

University of Wisconsin Residence Hall Honored Instructor 2024

Nominations are made by residents to honor instructors who have impacted them posi-
tively.

University of Rochester Outstanding Essay Award 2019

Awarded for the best essay written by a graduate student in philosophy.

Simon Fraser University Philosophy Graduate Research Paper Award 2016

Awarded for the best essay written by a graduate student in philosophy.

Teaching

Full Responsibility

Introductory AI & Data Ethics (UW - Madison)

Contemporary moral issues in AI and data ethics, including technical concepts, privacy, automation, and algorithms in medicine and law.

Introduction to Philosophy (×3 UW - Madison)

Introduction to classic and contemporary debates in philosophy, including metaphysics, epistemology, political philosophy, and ethics.

Ethics in Business (UW - Madison)

Contemporary moral issues in the workplace, including workplace governance, exploitation, whistle-blowing, and the future of work.

Introductory Ethics (UW - Madison)

Introduction to ethical theory, including major ethical theories, reasons to be moral, and moral luck.

Philosophy and the Sciences (UW - Madison)

Introduction to philosophy of science, including the scientific method, the problem of induction, the demarcation problem, and the value-free ideal.

Contemporary Moral Issues (UW - Madison)

Contemporary moral issues including charity and altruism, the duty to obey the law, democracy, AI, and oppression.

Teaching Assistant

Contemporary Moral Issues (UW - Madison)

Introduction to Philosophy (×2 UW - Madison)

Ethics in Business (UW - Madison)

Environmental Justice (University of Rochester)

Moral Problems (University of Rochester)

Philosophy of Artificial Intelligence (University of Rochester)

Philosophy of Science (Simon Fraser University)

Knowledge and Reality (×5 Simon Fraser University)

Moral Problems (×2 Simon Fraser University)

Research Assistantship

Project Assistant to Clinton Castro

Assisted Dr. Castro in developing an AI Ethics graduate seminar and assisted in his research on ethical issues regarding AI and children and liberal arguments for public libraries by finding and reviewing relevant academic literature.

Project Assistant to Elliott Sober

Assisted Dr. Sober in proofreading *The Philosophy of Evolutionary Theory* (Cambridge University Press, 2024), ensuring accuracy and clarity prior to publication, and assisted his research on gradualism in evolution and mathematical population genetics by finding and reviewing relevant academic literature.

Philosophy of Science Experiential Learning Program (SFU)

Assisted in developing a program to involve undergraduate students at Simon Fraser University in hands-on programming in the philosophy of science.

Solid Pathways Initiative with Sam Black (SFU)

Assisted in developing a critical thinking curriculum for Indigenous high school students in British Columbia by reviewing relevant classroom materials and exercises.

Service

Evolution Seminar Series Co-Chair

2025–present

Organized and hosted weekly seminars, including inviting speakers, organizing dinners, and handling AV gear for the J. F. Crow Institute for the Study of Evolution (Wisconsin Evolution).

Wisconsin Evolution Coordination Committee

2025–present

Helped to govern the J. F. Crow Institute for the Study of Evolution (Wisconsin Evolution), including meetings to organize budgets, speakers, journal clubs, and outreach activities.

Philosophy of Biology at Madison (PoBaM)

2026

Assisted Aja Watkins, Marina DiMarco, and Matt Haber in organizing and running the PoBaM conference.

Journal Referee

2026

Biology & Philosophy

UW-Madison Graduate Association of Philosophy - Director of Professional Service 2022–2023

Oversaw GAP funds, yearly elections, ensured compliance with bylaws, attended department meetings, acted as the voice of graduate students to faculty, and reported back to

constituents. In addition I oversaw the creation of a diversity and inclusion office.

SFU Graduate Student Society - Philosophy Caucus Representative 2015–2018

Oversaw budgetary and other matters of the Society and represented Philosophy grad students to the organization.

SFU Philosophy Teaching Assistant Handbook Co-Author 2018

Assisted in writing the SFU Philosophy Department's TA handbook, including chapters on the first day in the classroom and TAing philosophy of science.

SFU GRE Study Group Facilitator 2017

Organized and led workshops for philosophy students studying for the GRE.

FASS Forward Pilot Program 2017

Provided assistance to at-risk students through one-on-one meetings and mentoring in their first term at Simon Fraser University.

References

Elliott Sober

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William H. Bonsall Visiting Professor in the Humanities
Department of Philosophy
Stanford University and University of Wisconsin - Madison

Aja Watkins

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Mellon-Morgridge Professor and Assistant Professor
Department of Philosophy
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Hayley Clatterbuck

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Senior Researcher at Rethink Priorities and Visiting Associate Professor
Department of Philosophy
University of California Los Angeles

Clinton Castro

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Assistant Professor
The Information School
University of Wisconsin - Madison

Jesse Steinberg

jesse.steinberg@wisc.edu

Teaching Faculty and Assistant to the Chair
Department of Philosophy
University of Wisconsin - Madison

Graduate Coursework

Philosophy of Science

Parsimony (Elliott Sober, UW-Madison)
Actual Causation and General Explanation (Hayley Clatterbuck, Rochester)
Philosophy of Science (Hayley Clatterbuck, Rochester)
Darwin and Religion (Hayley Clatterbuck and H. Allen Orr, Rochester)
Philosophy of Mathematics (Zeynep Soysal, Rochester)
Explanation and Causation (Holly Andersen, SFU)
Laws of Nature (Alex Meehan, UW-Madison) *
Taxonomy and Classification (Aja Watkins, UW-Madison) *
Philosophy of Climate and Environmental Sciences (Aja Watkins, UW-Madison) *

Biology

Population Genetics (John Pool, Genetics, UW-Madison)
Adaptation (Chris Muir, Botany, UW-Madison)
Selection and Recombination (Aaron Ragsdale, Zoology, UW-Madison)
Microbial Astrobiology (Betul Kaçar, Microbiology, UW-Madison)
Foundations of Evolution (David Baum and Nathaniel Sharp, Botany,)

Historical Sciences

Early Modern Scientific Paperwork (Florence Hsia and Robin Rider, History, UW-Madison)
Infinity in Early Modern Philosophy (Anat Schechtman, UW-Madison)
History and Philosophy of Science (Holly Andersen, SFU) *

Epistemology

Epistemology After Aristotle (Peter Osorio, Rochester)
Theory of Knowledge (Earl Conee, Rochester)
Formal Epistemology (Nicolas Fillion, SFU)
Epistemology of Perception (Rosemary Twomey, SFU)
Social Epistemology (Endre Begby, SFU)
Epistemic Standards (Mike Titelbaum and Sophie Dandalet, UW-Madison) *
The Social Character of Knowledge (Jens Kipper, Rochester) *

Philosophy of Artificial Intelligence

Linguistic and Non-Linguistic Thought (Hayley Clatterbuck, UW-Madison)
Ethics of Algorithms (Jonathan Herrington, Rochester)
Philosophy of Artificial Intelligence (Jens Kipper, Rochester)

Logic

Hyperintensionality (Zeynep Soysal, Rochester)

Intermediate Logic (Zeynep Soysal, Rochester)

Mathematics of Morality (Nicolas Fillion, SFU)

Model Theory (Nicolas Fillion, SFU)

Philosophy as Analysis (Martin Hahn and Nicolas Fillion, SFU)

Other Courses

Spinoza's Ethics (Steve Nadler, UW-Madison)

Distributive Justice (Harry Brighouse, UW-Madison)

Shoulda, Coulda, Woulda (Jesse Steinberg, UW-Madison)

Social and Political Philosophy of Language (John McKay, UW-Madison)

Philosophy of Law (Jon Tresan, Rochester)

Grounding and Hyperintensional Metaphysics (Paul Audi, Rochester)

Hume and Reid (Alison Peterman, Rochester)

Philosophy of Philosophy (Philip Hanson, SFU)

Semantics and Pragmatics (Endre Begby, SFU)

Ownership: Self and World (Sam Black, SFU)

Meta-Normativity (Evan Tiffany, SFU)

Leibniz and Berkeley (Dai Heide, SFU)

Externalist Theories of Content (Martha Gibson, UW-Madison) *

* Indicates an audited course.

PHILOSOPHICAL ASPECTS OF GENETIC LOAD

MATTHEW J. MAXWELL

The past decade has seen a resurgence of “scientific” eugenics. One scientific concept used in defense of eugenics is genetic load, or the idea that variation imposes a “cost” on a population. The consequences, or costs, of genetic load for actual populations are, however, not well-understood. Genetic load is conceptually heterogeneous and in need of clarification. My aim in the dissertation is to provide that clarification and, in doing so, improve our understanding of natural selection, the sources of diversity, and cast doubt on eugenic arguments depending on genetic load.

In Chapter 1 I take on the argument from increasing genetic load to the necessity of eugenics. Eugenic arguments are frequently dismissed as racist pseudoscience. This may be true of many eugenic arguments, but the genetic load argument cannot be so easily dismissed. Many prominent geneticists, including fore-runners in the field, have argued that the risk to humans from genetic load may be existential.

I show, however, that the simple argument from relaxation of selection to fitness declines is invalid. An increase in frequency of deleterious alleles does not entail a decrease in fitness. Valid versions of the argument, on the other hand do not present an existential threat.

In Chapter 2 I examine two “paradoxes” of genetic load, which continue to be topics of debate in biology. One paradox argues that the amount of genetic diversity in natural populations is inexplicable, as natural selection should act to remove it. The second argues that the well-accepted Nearly-Neutral Theory predicts that the amount of load in natural populations is orders of magnitude too low. Application of this theory to humans, for example, predicts each reproducing female would need to have 16 offspring in order to maintain population size.

These paradoxes both trade on ambiguities in the concept of genetic load. Genetic load is commonly defined as the difference between the maximum fitness in a population and the mean fitness of the population. It is unclear, however, what the maximum fitness in a population is. This maximum might refer to the maximum possible fitness in a population, termed lag load, the maximum actual fitness in a population, termed substitution load, or the largely unstudied maximum likely fitness in a population. Each of these concepts is useful, but confusion among them has led to mistaken inferences.

Central to resolving these paradoxes is “soft selection,” or selection which is both frequency and density dependent, in which load and fitness-effects appear to come apart. Discussion of soft selection contributes to the philosophical literature on forms of selection, adding to the existing literature on frequency-dependent and balancing selection.

In Chapter 3 I address the effects of genetic load on the process of evolution. Large genetic loads are often thought to have significant negative effects on populations, often suggesting an increased chance of extinction via “mutational meltdown.” Increased numbers of mutations, however, also allow for increased probability of “evolutionary rescue,” i.e., avoiding extinction through adaptive evolution. But this may not be the only way that load can benefit a population. Soft selection may, in some circumstances, increase a population’s probability of survival while, in an apparent paradox, lowering the average fitness of the population. While some have invoked group selection to explain “altruistic” traits (traits that benefit others at a fitness cost

to the individual) such as these, this phenomenon is not a case of group selection.

In Chapter 4 I argue that genetic load holds lessons for thinking about laws in biology. The existence of laws in biology is controversial. One recent defense of a law in biology is Brandon and McShea's Zero-Force Evolutionary Law (ZFEL), which claims that in any evolutionary system there is a tendency for diversity and complexity to increase. Their proposal is not just a law; it is intended as a "gestalt shift," which implies that diversity and complexity do not, in general, stand in need of explanation. Where others have concluded that research into the sources of diversity supports ZFEL as a law, I argue that this same research undermines it as a gestalt shift.